Insurance Claims Frequency and Severity

McQueen Insurances

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1 Executive Summary

2 Data Preparation

```
setwd("C:/Users/MathieuDemarets/Documents/UCL Certificat/CodeRepositories/LACTU2150/project")
load("../data/mtpl_be.rda")
data <- mtpl_be
```

2.1 Data Exploration

2.1.1 Initial Data Overview

```
str(data)
```

```
## 'data.frame':    163231 obs. of  18 variables:
##  $ ID      : int  1 2 3 4 5 6 7 8 9 10 ...
##  $ NCLAIMS : int  1 0 0 0 1 0 1 0 0 0 ...
##  $ AMOUNT  : num  1618 0 0 0 156 ...
##  $ AVG     : num  1618 NA NA NA 156 ...
##  $ EXP     : num  1 1 1 1 0.0466 ...
##  $ COVERAGE: chr  "TPL" "PO" "TPL" "TPL" ...
##  $ FUEL    : chr  "gasoline" "gasoline" "diesel" "gasoline" ...
##  $ USE     : chr  "private" "private" "private" "private" ...
##  $ FLEET   : chr  "N" "N" "N" "N" ...
##  $ SEX     : chr  "male" "female" "male" "male" ...
##  $ AGEPH   : int  50 64 60 77 28 26 26 58 59 34 ...
##  $ BM      : int  5 5 0 0 9 11 11 11 0 7 ...
##  $ AGEC    : int  12 3 10 15 7 12 8 14 3 6 ...
##  $ POWER   : int  77 66 70 57 70 70 55 47 98 74 ...
##  $ PC      : int  1000 1000 1000 1000 1000 1000 1000 1000 1000 1000 ...
##  $ TOWN    : chr  "BRUSSEL" "BRUSSEL" "BRUSSEL" "BRUSSEL" ...
##  $ LONG    : num  4.36 4.36 4.36 4.36 4.36 ...
##  $ LAT     : num  50.8 50.8 50.8 50.8 50.8 ...
```

```
summary(data)
```

```
##        ID            NCLAIMS          AMOUNT               AVG          
##  Min.   :     1   Min.   :0.000   Min.   :      0.0   Min.   :2.000e-02  
##  1st Qu.: 40809   1st Qu.:0.000   1st Qu.:      0.0   1st Qu.:1.428e+02  
##  Median : 81616   Median :0.000   Median :      0.0   Median :5.298e+02  
##  Mean   : 81616   Mean   :0.124   Mean   :    200.8   Mean   :1.626e+03  
##  3rd Qu.:122424   3rd Qu.:0.000   3rd Qu.:      0.0   3rd Qu.:1.426e+03  
##  Max.   :163231   Max.   :5.000   Max.   :1989567.9   Max.   :1.990e+06  
##                                                       NAs    :144936     
##       EXP              COVERAGE             FUEL               USE          
##  Min.   :0.00274   Length   :163231   Length   :163231   Length   :163231  
##  1st Qu.:1.00000   N.unique :      3   N.unique :      2   N.unique :      2  
##  Median :1.00000   N.blank  :      0   N.blank  :      0   N.blank  :      0  
##  Mean   :0.88974   Min.nchar:      2   Min.nchar:      6   Min.nchar:      4  
##  3rd Qu.:1.00000   Max.nchar:      3   Max.nchar:      8   Max.nchar:      7  
##  Max.   :1.00000                                                            
##                                                                             
##       FLEET               SEX              AGEPH           BM        
##  Length   :163231   Length   :163231   Min.   :18   Min.   : 0.000  
```

```
##  N.unique :     2   N.unique :     2   1st Qu.:35   1st Qu.: 0.000
##  N.blank  :     0   N.blank  :     0   Median :46   Median : 1.000
##  Min.nchar:     1   Min.nchar:     4   Mean   :47   Mean   : 3.269
##  Max.nchar:     1   Max.nchar:     6   3rd Qu.:58   3rd Qu.: 6.000
##                                        Max.   :95   Max.   :22.000
##
##       AGEC            POWER           PC              TOWN
##  Min.   : 0.000   Min.   : 10   Min.   :1000   Length   :163231
##  1st Qu.: 4.000   1st Qu.: 43   1st Qu.:2900   N.unique :   578
##  Median : 7.000   Median : 53   Median :5030   N.blank  :     0
##  Mean   : 7.375   Mean   : 56   Mean   :5271   Min.nchar:     2
##  3rd Qu.:10.000   3rd Qu.: 66   3rd Qu.:7500   Max.nchar:     8
##  Max.   :48.000   Max.   :243   Max.   :9990
##
##       LONG            LAT
##  Min.   :2.578   Min.   :49.54
##  1st Qu.:3.879   1st Qu.:50.52
##  Median :4.387   Median :50.77
##  Mean   :4.407   Mean   :50.76
##  3rd Qu.:4.874   3rd Qu.:50.99
##  Max.   :6.306   Max.   :51.45
##
```

From this initial data loading, we can already notice that some columns have been asigned the wrong type: - *COVERAGE*, *FUEL*, *USE*, *FLEET*, *SEX*, and *TOWN* are set as character variables and should be transformed into factor variables. - *PC* is wrongly defined as an integer and should be transformed into a factor variable. - *BM* is defined as bonus-malus level. A higher level indicates a worse claim history but it is not specified that the difference between levels is uniform. Therefore, it should be treated as an ordinal factor variable.

We will also make sure that the reference levels are correctly set for the factor variables: - Most frequent level for unordered factor variables (e.g., COVERAGE, FUEL, USE, FLEET, SEX, TOWN) - Lowest level for ordinal factor variables (e.g., BM)

```
for (col in c("COVERAGE", "FUEL", "USE", "FLEET", "SEX", "TOWN", "PC")) {
    data[[col]] <- as.factor(data[[col]]) %>% fct_infreq()
}
data[["BM"]] <- factor(data[["BM"]], ordered = TRUE)
```

We can then assert that the categorical variables (ordered or unordered) have the correct reference levels.

```
assert_ref_level <- function(data, cols) {
    for (col in cols) {
        if (!is.factor(data[[col]])) stop(paste(col, "is not a factor variable"))
        if (is.ordered(data[[col]])) {
            if (as.numeric(levels(data[[col]]))[1] != min(as.numeric(as.character(data[[col]]))
            # Pay attention to turn labels as numeric and not the factor internal codes which
```

```
                stop(paste("Reference level for", col, "is not the lowest level"))
            }
        } else {
            if (
                levels(data[[col]])[1] !=
                    fct_count(data[[col]], sort = TRUE)$f[1]
            ) {
                stop(
                    paste(
                        "Reference level for", col,
                        "is not the most frequent level"
                    )
                )
            }
    }
    }
}
assert_ref_level(data,  c("COVERAGE", "FUEL", "USE", "FLEET", "SEX", "TOWN", "PC", "BM"))
```

2.1.2 Statistical Approach

2.1.3 Graphical Approach

2.1.4 Machine Learning Approach

2.2 Feature Engineering

2.2.1 Geospatial Features

2.2.2 Polynomial Features

2.2.3 Interaction Features

3 Modelling

3.1 Validation Framework

3.1.1 Performance Metrics

3.1.2 Statistical Tests

3.1.3 Cross-Validation

3.2 Frequency Modelling

3.2.1 Naive and Full Models

3.2.2 GLM Models

3.2.3 GAM Models

3.2.4 Hurdle Models

3.2.5 Final Frequency Model

3.3 Severity Modelling

3.3.1 GLM Models

3.3.2 Final Severity Model

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